

YOUSSEF GAMAL SALEM

AI Engineering Student | Software Engineering / AI & ML Intern Candidate

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PROFILE

Fourth-year AI Engineering student at New Ismailia National University (NINU), expected to graduate in 2028, with hands-on academic project experience in machine learning, computer vision, databases, embedded systems, IoT, and networking. Strong focus on Python-based AI development, CNNs, image preprocessing, hyperparameter optimization, and practical system implementation. Seeking a software engineering or AI/ML internship to apply technical skills to real-world products and continue developing as an engineer.

TECHNICAL SKILLS

Programming	Python, C, SQL
AI / ML	Machine Learning, Deep Learning, CNNs, Computer Vision, Image Preprocessing, Data Augmentation, Hyperparameter Optimization, Particle Swarm Optimization (PSO)
Frameworks / Libraries	TensorFlow, Keras, scikit-learn, PySwarms
Databases / App	PostgreSQL, Neon, Streamlit
Embedded / IoT	ESP32, AT89C51, ADC0804, LM35, LCD, Keil, Proteus
Networking	Cisco Packet Tracer, Routers, Switches, Access Points, DHCP

SELECTED PROJECTS

Brain Tumor MRI Classification | Python, TensorFlow/Keras, scikit-learn, PySwarms

- Built a CNN-based multi-class classifier for glioma, meningioma, pituitary, and no-tumor MRI images.
- Applied image preprocessing including grayscale conversion, cropping/ROI focus, normalization, histogram equalization, and Gaussian denoising.
- Implemented augmentation and Particle Swarm Optimization (PSO) to search CNN hyperparameters and improve validation performance.

Citizen Registry Database | PostgreSQL, Neon, Streamlit

- Designed a citizen registry database with relational structure, SQL queries, ER modeling, and database triggers.
- Developed a Streamlit frontend connected to a PostgreSQL database hosted on Neon for practical data interaction.

Digital Temperature Sensor | C, Keil, Proteus, AT89C51

- Developed a temperature-monitoring system using AT89C51, LM35DZ, ADC0804, LCD, 74LS373 latch, and external 27C64 EPROM.
- Implemented the microcontroller/EPROM interface and simulated the complete system in Proteus.

Smart Home IoT System | ESP32, 4-Channel Relay, Voice Control

- Built an ESP32-based smart-home control concept using a 4-channel relay and explored Alexa, Google Assistant, and IFTTT integration for voice control.

Computer Networking Design | Cisco Packet Tracer

- Designed a university-lab network using routers, switches, access points, and DHCP addressing for multiple labs and client devices.

EDUCATION

New Ismailia National University (NINU) — B.Sc. in AI Engineering

Fourth Year | Expected Graduation: 2028 | Ismailia, Egypt

CAREER OBJECTIVE

Seeking an internship in Software Engineering or AI/ML where I can contribute to development projects involving Python, machine learning, backend/data systems, or applied AI while gaining professional engineering experience.